FACULTY OF COMPUTING

SECP3843 - 02 SPECIAL TOPIC IN DATA ENGINEERING

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TUTORIAL MS AZURE REPORT

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TABLE OF CONTENT

1.0 Project Overview

The primary objective of this project is to bridge the gap in customer demographic intelligence for a retail-oriented business. Currently, the company’s data is siloed in an on-premises SQL database, making it difficult for stakeholders to analyze how gender distribution affects purchasing behavior across different product categories.

1.1 Problem Statement

Stakeholders lack visibility into:

1. Total products sold and revenue distribution by gender.
2. The correlation between specific product categories and customer demographics.
3. The ability to perform self-service, date-based filtering for real-time decision-making.

1.2 Business Value

By migrating this data to a cloud-based environment, the project enables a comprehensive KPI dashboard that empowers leadership to optimize marketing strategies and inventory management based on high-fidelity customer insights.

2.0 Description on the solution

The Azure Data Ecosystem was used to create an end-to-end automated data pipeline in order to address this. The solution transfers data from on-premises to a state suited for visualization using a contemporary data engineering paradigm.

2.1 Technical Design

The four primary pillars of the solution are as follows:

1. Data Ingestion (Bronze): We load raw data into Azure Data Lake Storage (ADLS) Gen2 after extracting it from the on-premises SQL Server using Azure Data Factory. This maintains the original data format and acts as our "Bronze" layer.
2. Data Transformation (Silver): The raw data is cleaned using Azure Databricks (Spark). This entails managing missing values, standardizing schemas, and improving data types to guarantee superior quality.
3. Data Enrichment (Gold): Specifically designed for reporting, refined data is combined into "Gold" tables. In this step, sales transactions and customer demographics are combined to provide the desired KPIs (e.g., Sales by Gender).
4. Automation & Visualization: ADF pipelines that are scheduled to run every day orchestrate the entire workflow. A Power BI dashboard that offers interactive filters for gender, category, and date ranges is linked to the final output.

2.2 Important Aspects of the Solution

Scalability: Based on cloud-native technologies that can manage growing amounts of data.

Reliability: Without the need for human involvement, automated scheduling guarantees that stakeholders always have the most recent information.

Security: To safeguard private client data, secure authentication was implemented utilizing Access Tokens and linked services.

3.0 Technology Use

In this project, a multi-layered Azure ecosystem was used to guarantee a scalable and secure data pipeline.

3.1 Azure Data Lake storage (ADLS) Gen2: This was the main storage layer in the Medallion architecture (Bronze, Silver, and Gold). DFS endpoint played a vital role in supporting hierarchical namespaces to enable Synapse and Databricks to list and process Delta Lake transaction logs efficiently.

3.2 Azure Data Factory (ADF): It served as the control hub. It used dynamic elements like Get Metadata and ForEach Loops to automate the process of ingesting and processing multiple tables, which greatly minimized the amount of manual configuration.

3.3 Azure Synapse Analytics (Serverless SQL Pool): Enabled data virtualization. With the use of Database Scoped Credentials (via SAS tokens) and External Data Sources, a secure bridge was created to query Delta tables out of the storage layer without transferring the data.

3.4 Azure Databricks: Applied at the transformation stage, which involves processing raw data using Spark to clean and refine it into the high-quality Delta format needed to be loaded into the Gold layer and further analytics.

3.5 Azure SQL Database: Was used as the main data source, and it emulated an on-premise SQL setup to show how structured operational data can be ingested into the cloud.

3.6 Azure Key Vault: Secured the pipeline by offering a central repository of sensitive credentials and SAS tokens.

3.7 Azure Active Directory (Entra ID): Identity and Access Management, which makes sure that only authorized users and services are able to access the data assets of the pipeline.

4.0 Data Pipeline Architecture

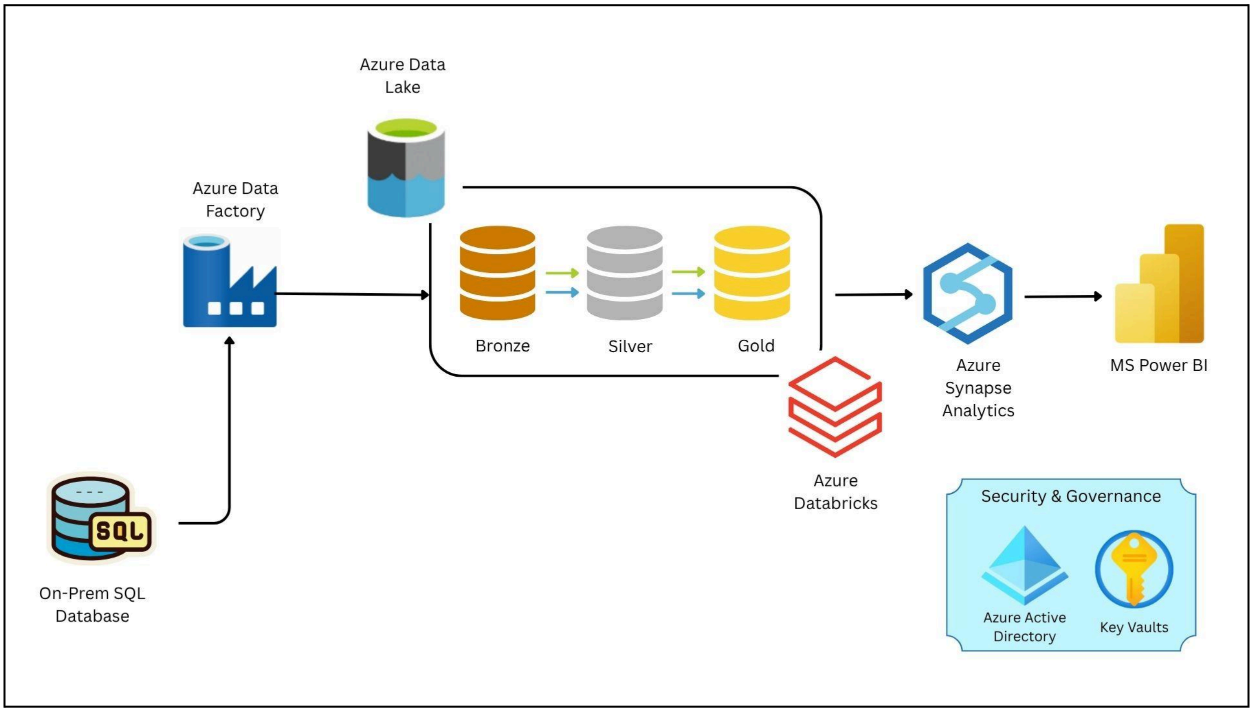

Figure 4.0

Figure 4.0 shows a data pipeline architecture from end-to-end. It consists of data sources, ingestion, transformation, storage, to analytics layers by using all the technology stack mentioned above.

4.1 Data Source

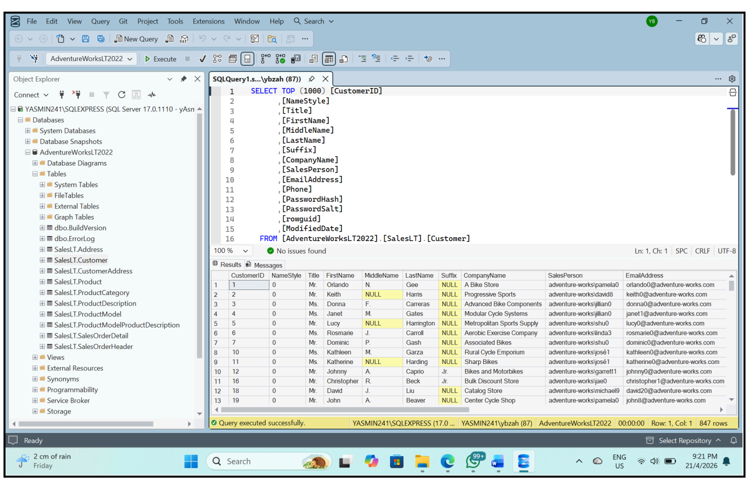

Figure 4.1 Data AdventureWorksLT2022

Figure 4.1 shows data AdventureWorksLT2022 that is downloaded from Microsoft Learn. Then, the data files were extracted into SQL Server Management Studio for viewing the dataset, and making any adjustment into the pipeline.

4.2 Data Ingestion

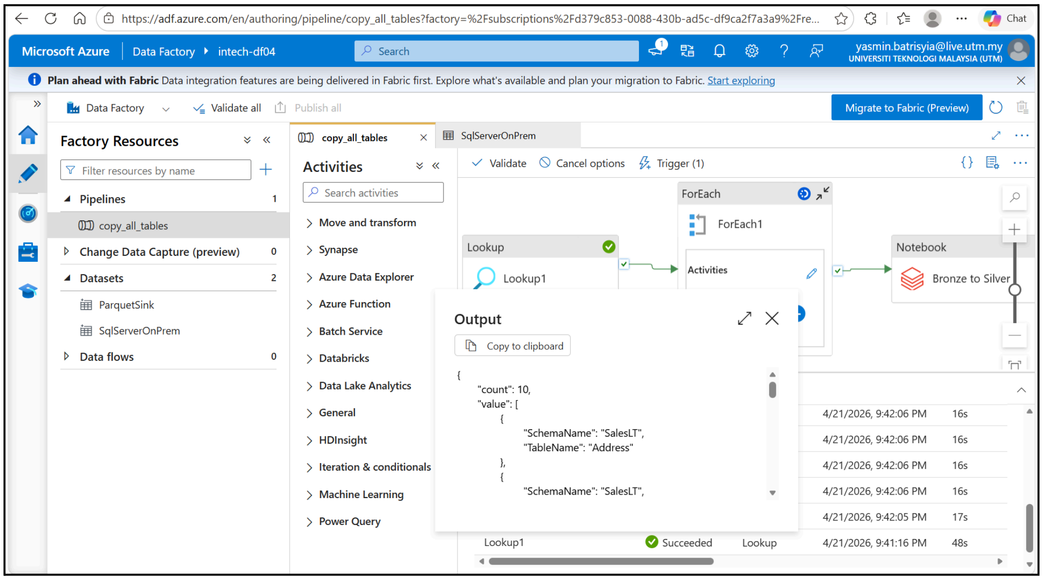

Figure 4.2.1 Data Ingestion Lookup

In this stage as shown in Figure 4.2.1, the data from an on premise sql database has been extracted and loaded into Azure Data Lake Storage (ADLS) by using Azure Data Factory (ADF).

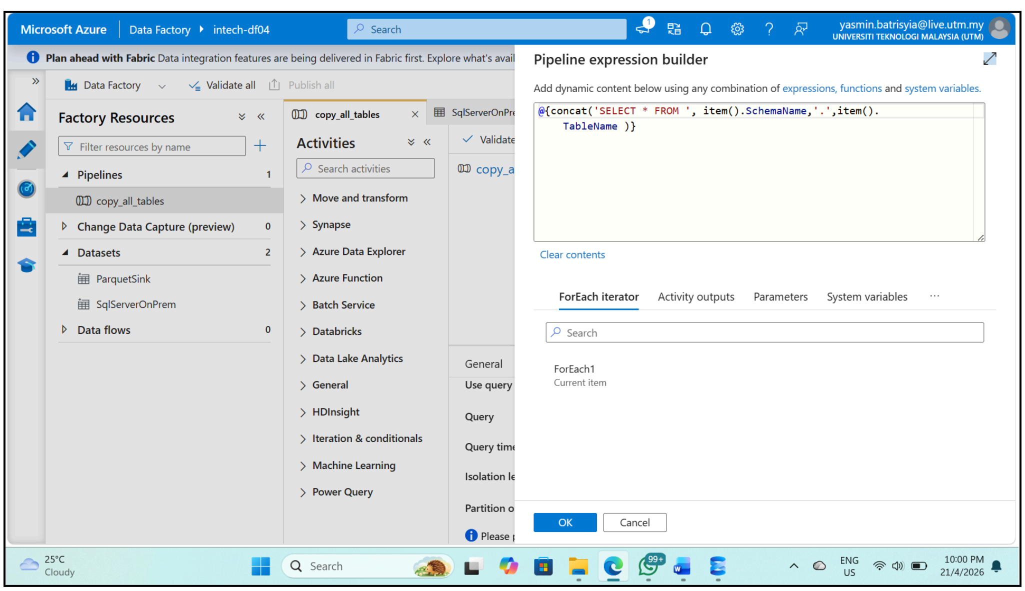

Figure 4.2.2 Data Ingestion For Each Activity

Then, the data dynamic query is designed to pull from the source table. By using the @concat function, Azure Data Factory merges the static SQL command 'SELECT * FROM ' with two dynamic properties which are SchemaName and TableName. Then, extracted from the current item() in the loop. The inclusion of a literal dot '.' between these properties ensures the final output follows the standard SQL "dot notation", allowing a single activity to automatically adapt and query different tables without any manual configuration changes.

4.3 Data Transformation

Azure Databricks is used to clean and transform the data. The data is then organized into the 3 layers which are bronze, silver and gold for raw, cleansed, and aggregated data respectively.

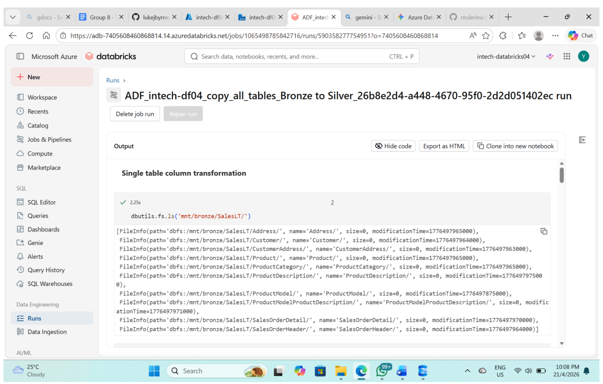

Figure 4.3.1

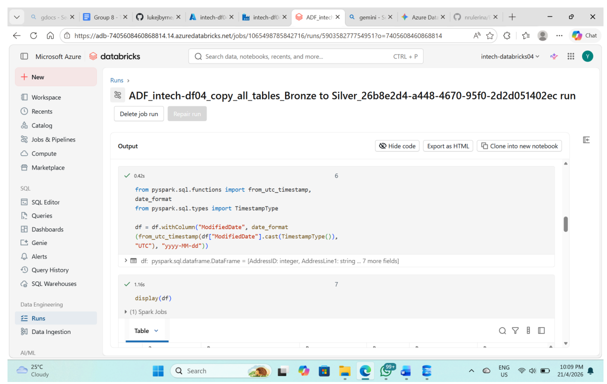

Figure 4.3.2

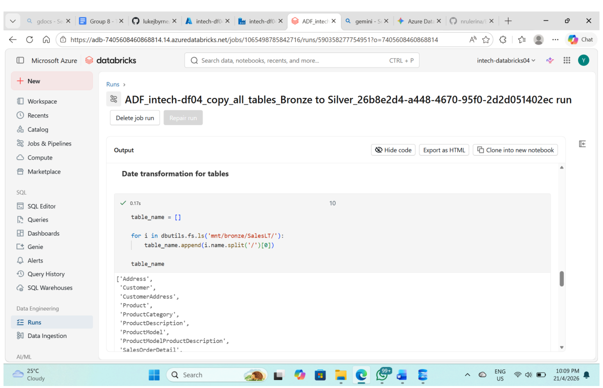

Figure 4.3.3

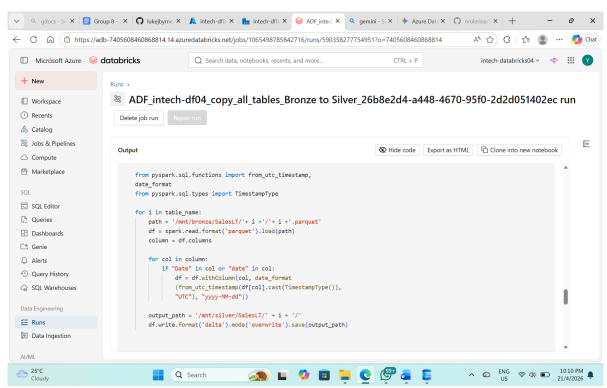

Figure 4.3.4

Figure 4.3.1 to 4.3.4 showed how data was cleaned such as modifying the dates into more readable form and transforming it into the silver layer.

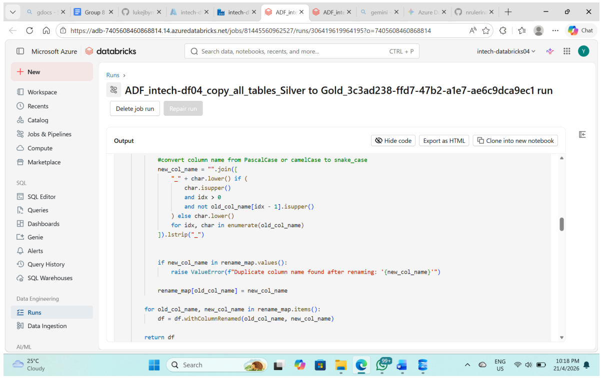

Figure 4.3.5

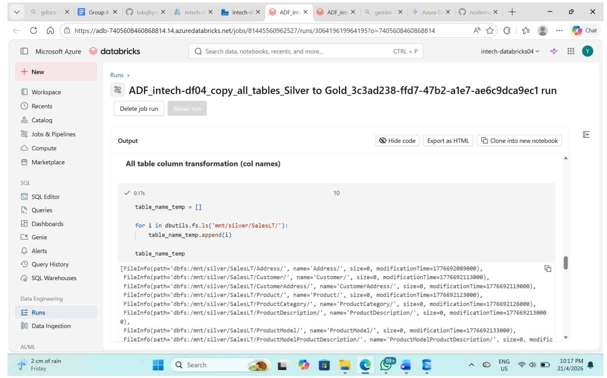

Figure 4.3.6

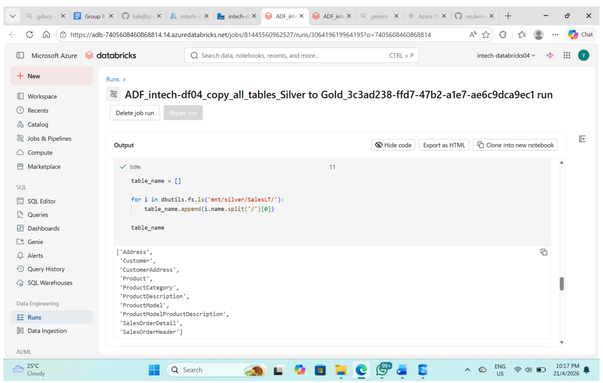

Figure 4.3.7

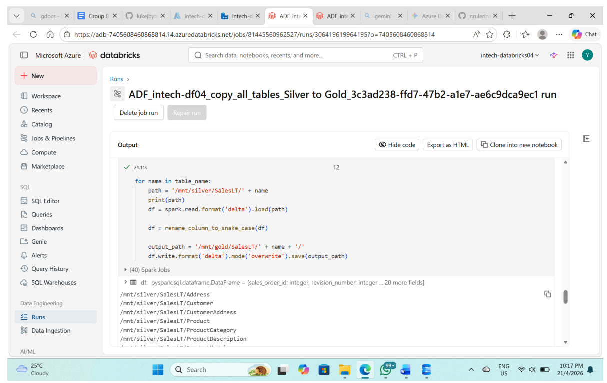

Figure 4.3.8

Next, as shown in Figure 4.3.5 to 4.3.8, the adjustments that have been done in this layer are the column name from PascalCase or camelCase was converted to snake_case in a PySpark DataFrame. Moreover, the new column name was separated by an underscore for clearer understanding of the column name in the future analysis. Then, the current data was transferred into the gold layer.

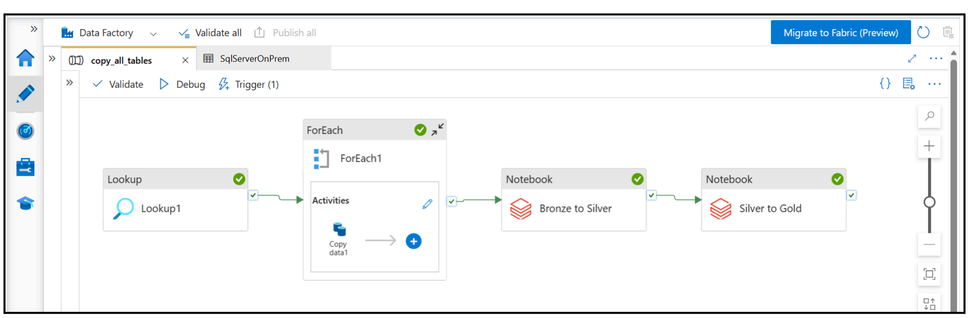

Figure 4.3.9

Figure 4.3.9 shows that all the data is successfully transformed from bronze layer to gold layer. It means that all the data is analytical ready to use.

4.4 Data Storage

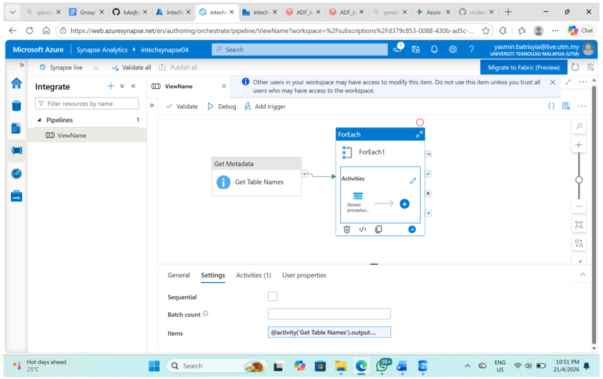

Figure 4.4.1

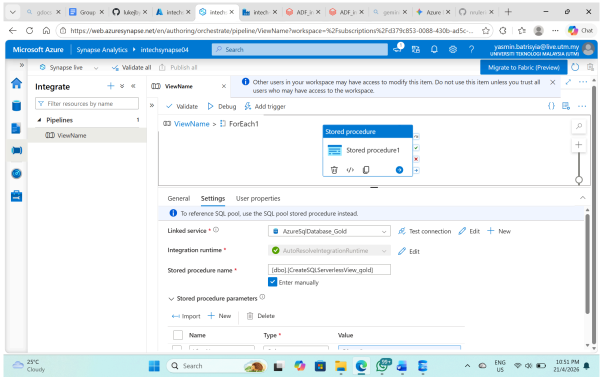

Figure 4.4.2

Figure 4.4.1 and 4.4.2 would act as a data warehouse in Azure Synapse. This is because the data that has been stored is from the gold layer, and the data is ready for high performance querying.

4.5 Analytics layer

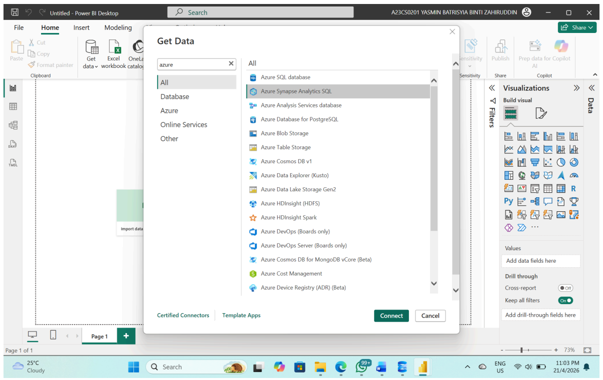

Figure 4.5.1

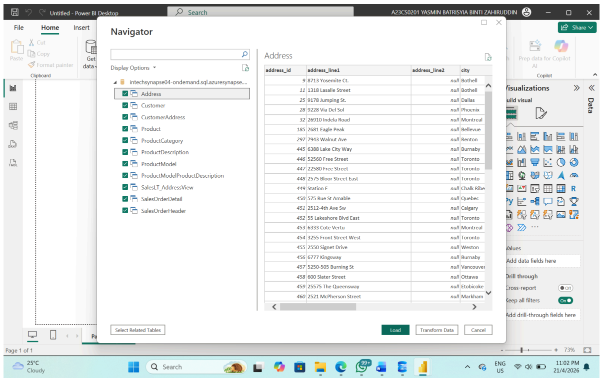

Figure 4.5.2

To perform analysis within Power BI, Azure Synapse is connected to the platform for data retrieval. Subsequently, the selected tables are loaded into Power BI, enabling the creation of the dashboard.

5.0 MS Power BI dashboard

The dashboard is designed to transform raw operational data into actionable business intelligence through professional visualizations and clear insights.

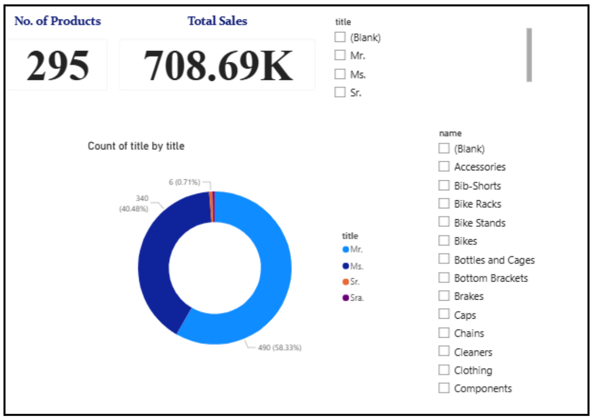

***Figure 5.1** Dashboard of product purchases*

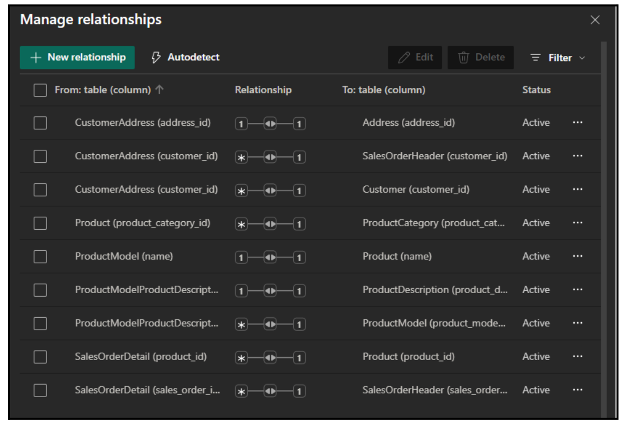

***Figure 5.2** Existing relationships*

***Figure 5.3** Model view in Power BI*

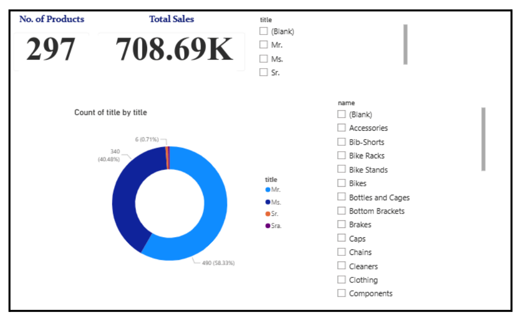

***Figure 5.4** Updated dashboard*

Meaningful Visualizations & Insights:

- **KPI Card**: The card visual showing a Total Product Count of 295 provides an immediate baseline for the scale of operations. This allows operations managers to quickly verify inventory health against expected quotas.
- **Customer Demographic Donut Chart**: This chart reveals that approximately 58.3% of customers are identified as 'Mr.', indicating a male-dominant customer base. This insight

allows the business to tailor marketing strategies and product recommendations toward this specific demographic.

- **Interactive Category Slicer**: The slicer enables dynamic cross-filtering by product category (e.g., Bikes, Accessories). By selecting 'Bike Racks', the business can observe real-time shifts in customer titles, identifying which niche products appeal most to different demographic segments.

Professional Design & Modeling:

- **Data Modeling**: To support these insights, a robust Many-to-One (*:1) relationship chain was implemented, ensuring data integrity across the entire schema.
- **Aesthetic Clarity**: By removing redundant category labels and centering callout values, the dashboard maintains a high-quality "storytelling" feel that prioritizes data readability.

6.0 Reflection

NURUL ADRIANA BINTI KAMAL JEFRI (A23CS0258)

Doing this project gave me a profound understanding of the intricacies of configuring a cloud-based Medallion architecture in MS Azure. One of my biggest issues was finding that the technical handshake between services could be particularly problematic, especially fixing permission-related errors of the form directory that cannot be listed by properly configuring DFS endpoints and filtering system-generated folders in Azure Data Factory. Overcoming these obstacles helped me to understand the essential role of safe credential management with the SAS tokens and the effectiveness of working with Stored Procedures to automatize data virtualization in Azure Synapse. In my team of three students, we shared our tasks, but between pipeline development and dashboard design, I had to be in constant contact with my teammates to make sure that the Gold layer schema matched the needs of Power BI modeling in every detail. This collaboration strengthened both my technical modeling skills and my ability to present data as a compelling business story.

SYARIFAH DANIA BINTI SYED ABU BAKAR (A23CS0183)

In this project, I successfully turned unprocessed on-premises SQL data into useful business intelligence by utilizing the Azure ecosystem to build a strong data engineering pipeline. I coordinated the data flow via Azure Databricks' Bronze, Silver, and Gold levels using a modular Medallion Architecture, guaranteeing data integrity through methodical cleaning and aggregation. Overcoming technical obstacles, such setting up secure authentication using Azure Access Tokens and resolving Spark execution issues, helped me gain a deeper grasp of cloud architecture and the significance of creating idempotent, scalable workflows. In the end, this experience helped close the gap between theoretical data modeling and real-world cloud deployment, producing an automated system that can give stakeholders vital information on client demographics and sales success.

YASMIN BATRISYIA BINTI ZAHIRUDDIN (A23CS0201)

Through this project, I gained a lot of experience in developing an end-to-end data pipeline within the Microsoft Azure ecosystem. I successfully navigated the entire data lifecycle, including environment setup, data ingestion, transformation, loading, and reporting.

Furthermore, I implemented automation and monitoring protocols and conducted thorough end-to-end testing to ensure system reliability. It is not an easy journey as I've faced several technical challenges, such as troubleshooting connectivity issues between the on-premises database and Azure Data Factory, resolving configuration errors in Databricks, and properly setting up Azure Synapse Analytics. Despite these hurdles, I successfully resolved each problem, resulting in a fully functional data pipeline. I really valued the opportunity to transform the theoretical concepts of Medallion Architecture into a practical, working model by following the Youtube tutorials. This project has been a significant learning milestone for me, and it has highly encouraged me to continue exploring and building functional data pipelines in the future.